

Part 1: Setting up the Server and Building the Docker Image

1. Initial Setup on AWS EC2 (Server Instance)

Connect to your AWS EC2 instance named *****My server***** and run the following commands:

```
``bash
```

```
# Switch to root user
```

```
>>sudo su
```

```
# Update the package index
```

```
>>apt update
```

```
# Install Docker
```

```
>>apt install docker.io -y
```

```
...
```

2. Creating the Dockerfile

Open a new file named Dockerfile using the Vim editor:

```
``bash
```

```
>>vi Dockerfile
```

```
...
```

*Press ****i**** to enter ****INSERT**** mode, paste the following configuration, then press ****ESC**** and type ****:wq**** to save and exit.*

```
``dockerfile
```

```
>>FROM ubuntu
```

```
>>RUN apt update
```

```
>>RUN apt install apache2 -y
```

```
>>RUN apt install git -y
```

```
>>RUN rm -rf /var/www/html/*
```

```
>>RUN git clone https://github.com/akshu20791/apachewebsite /var/www/html/
```

```
>>CMD ["apachectl", "-D", "FOREGROUND"]
```

```
...
```

3. Build the Docker Image

```
```bash
>>docker build -t image1:V1 .
```

```
Verify the image was created
docker images
```

```
...
```

## ## Part 2: Pushing the Image to Docker Hub

### ### 1. Generate a Personal Access Token (PAT) on Docker Hub

1. Open Docker Hub and sign in.
2. Go to **Profile** >> **Account Settings** >> **Security**.
3. Click **New Access Token**.
4. Give it a name in the **Access token description**.
5. Set access permissions to **Read, Write, Delete**.
6. Click **Generate**.
7. Copy the generated token/password

```
.
```

### ### 2. Authenticate and Push from the EC2 Terminal

```
```bash
# Log in to your Docker Hub account
>>docker login -u amulyayarlagadda
```

```
# When prompted for a password, paste the Personal Access Token you copied
```

```
# Tag your local image with your Docker Hub username
>>docker tag image1:V1 amulyayarlagadda/image1:V1
```

```
# Push the image to Docker Hub
>>docker push amulyayarlagadda/image1:V1
```

```
...
```

Part 3: Deploying on the Client Machine

1. Initial Setup on the Client EC2 Instance

Create and open another EC2 instance to act as your **Client machine**.

```
```bash
Switch to root user
>>sudo su

Update package index and install Docker
>>apt update
>>apt install docker.io -y

Pull the image from Docker Hub
>>docker pull amulyayarlagadda/image1:V1

Verify the image downloaded successfully
>>docker images
```

...

### ### 2. Run the Container

Run the container interactively, mapping port 2605 on the host to port 80 inside the container:

```
```bash
>>docker run -it -p 2605:80 --name client amulyayarlagadda/image1:V1 /bin/bash
```

...

3. Update AWS Security Group Settings

To allow external traffic to reach your application, update your Client EC2 instance's firewall settings in the AWS Console:

1. Click on your **Instance ID**.
2. Go to **Security** >> **Security groups**.
3. Select **Inbound rules** and click **Edit inbound rules**.
4. Click **Add rule** and set:
 - * **Type:** Custom TCP
 - * **Port Range:** 2605
 - * **Source:** Anywhere-IPv4 (0.0.0.0/0)
5. Click **Save rules**.

4. Start Apache Inside the Container & Verify

If Apache doesn't start automatically inside your interactive container terminal, check its status and start it:

```
```bash
>>service apache2 status
>>service apache2 start
```

...

### ### 5. Access Your Website

Open a web browser and navigate to:

```
```text
http://<client_public_IP>:2605
```

...